

are highly correlated. This increased volatility can aggravate investors' poor timing decisions.

Using the same data, Kumar (2009) discovered that investors prefer to hold underperforming lottery-style stocks that are low-priced and have high volatility or high skewness.²² The author determined that the typical investor could have improved his or her performance by 2.8% annually if he or she had simply replaced the lottery component of his or her portfolio with the nonlottery component.

Barber and Odean (2000) noted that the net stock market returns earned by average households lagged reasonable benchmarks by economically and statistically significant amounts. Individual investors turn over 80% of their portfolios annually. As we saw in [Chapter 4](#), investors trade too much due to overconfidence and the disposition effect that makes them greedy and fearful at the wrong times. Warren Buffett recommends doing the exact opposite of this: be greedy when others are fearful and fearful when others are greedy.

Weber et al. (2014), using data from an online European discount broker from 1999 through 2011, found that 5,000 individual European investors earned slightly negative risk-adjusted returns. Like their U.S. counterparts, European investors held only a few stocks, and these were often highly correlated. European investors also had a lottery mentality by preferring low-priced stocks with high idiosyncratic volatility or high idiosyncratic skewness. Eliminating some of these investor behaviors could have improved the average investor's annual returns by 4% for underdiversification and by 3% for lottery stock preferences.

Individual investors are also at a disadvantage in terms of information and expertise. When Michael Steinhardt, the renowned hedge fund manager, was asked what the most important thing an average investor could learn from him, he replied, "I'm their competition."

Warren Buffett said, "Investing is simple but not easy." In summary, individual investors tend to

- Respond overemotionally to market volatility
- Hold high-volatility, lottery-style stocks and underdiversified portfolios
- Be overconfident and overtrade their stock holdings